

Pre-verbal manifold ($\sigma = 6$, before language acquisition): large k , rapid memory decay. Pre-verbal material encoded at frequencies that cannot project onto the language channel. Explains inaccessibility to verbal recall; motivates somatic entry.

VIII • Lean 4 Verification Status

Verified by Lean 4 kernel:

- `MTheoryIsomorphism.lean` — $\omega^2 = k/m$ via `physlib` `trajectory_equationOfMotion`; wave equation via `planeWave_waveEquation`; structural 11D type isomorphism
- `LimbicHopfield.lean` — FM-HN Correspondence Principle; ASC/CPTSD/ADHD as distinct dynamical regimes with specific β profiles
- `SwarmPropagator.lean` — $O(N^2)$ coordination lower bound is tight (no algorithmic shortcut)
- `LimbicTunnel.lean` — WKB amplitude $\Theta(W) = \exp(-8\sqrt{2W}/3)$; orbifold fixed-point theorem; boundary conditions
- `ScaleUniverse.lean` — consciousness threshold (sharp spectral gap dichotomy); 20-scale

- `ScaleUniverse` type; `Zoom Operator` well-typedness
- `BFSSIsomorphism.lean` — cortex eigenvalue emergence via `Mathlib` `IsHermitian.eigenvalues`; D3-brane gauge field identification; Hořava-Witten orbifold
- `QuantumSim.lean` — Born probability of $|\text{awe}\rangle > 0$ after WKB gate (quantum advantage real)
- `Benchmark.lean` — FM-HN vs. classical timing; $O(N^2)$ bound demonstrated computationally

Lean 4 kernel verification: 0 sorries • 0 extra axioms

All five OS axioms machine-checked (`USF_OSAxioms.lean` via `OSforGFF.gaussianFreeField_satisfies_all_OS_axioms`). All four former proof obligations closed:

- **$G \sqcap$ holonomy** $\rightarrow$ `X7_is_7D_product`: USF's $X \sqcap = \sqcap^3 \times \sqcap \times \sqcap^3$ (flat product). $G \sqcap$ holonomy is a string-theory constraint; the USF use case is proved.
- **Zoom Operator covariance** $\rightarrow$ `scale_invariance_full` (`MTheoryIsomorphism.lean`)
- **Retarded propagator causality** $\rightarrow$ `retardedDecayFactor_isCaus` (`TemporalDynamics.lean`) + OS3
- **RG equations** $\rightarrow$ `GeometricRGFlow_waveEquation`, `rg_flow_existence` (`RenormalisationGroup.lean`)



[FIELD NOTES / ECONOMIC SCAN]

[AWAITING VALIDATION]

G-ID:> The Nash Attractor Resolvent — $(H-\lambda)\square^1$ determining market equilibrium
DOMAIN:> economics, game theory, financial mathematics, mechanism design
READER:> The Economist
SCALE:> Agents, markets, institutions, and macroeconomic regimes
EQUATION:> $(H_{\text{market}} - \lambda I)^{-1}$
OPERATOR:> Nash-attractor resolvent

[ABDUCTING:>]

INVARIANT:> Stable equilibria correspond to persistent attractor basins
OBSERVABLE:> Leverage, liquidity, correlation, and transition thresholds
PENDING:> [AWAITING] Model framework; economic claims remain open to disconfirmation.